

EXPERIMENT NO 8

Aim: Access and validate form fields using events; design a gym admission form with live input checks.

Requirements: A computer with a web browser, Visual Studio Code (or any text editor), HTML, CSS, and JavaScript.

Theory:

Form validation is the process of checking whether the data entered by a user is correct, complete, and in the required format before submitting the form. In web applications, events are used to detect user actions such as typing, focusing on a field, changing a value, or submitting the form.

Common events used for live form validation are:

- input – triggered whenever the user enters or modifies text.
- change – triggered when the value of a field is changed.
- focus – occurs when the user selects a form field.
- blur – occurs when the user leaves a form field.
- submit – occurs when the user submits the form.

JavaScript can access form fields using methods such as `getElementById()` or `querySelector()`. Validation rules can then be applied to fields such as name, email, phone number, age, membership type, and password.

For a Gym Admission Form, live validation can be used to:

- Ensure the member's name is not empty.
- Check that the age is within an acceptable range.
- Validate the phone number.
- Validate the email address.
- Ensure a membership plan is selected.
- Display error messages immediately while the user is entering data.

This improves user experience and prevents invalid information from being submitted.

Algorithm:

1. Start
2. Create a Gym Admission Form containing fields such as:
 - Full Name
 - Age
 - Email
 - Phone Number
 - Membership Plan
 - Address
3. Access each form field using JavaScript.
4. Attach event listeners such as input, blur, change, and submit.
5. When the user enters a name:
 - Check whether it is empty.
 - Display an error if the field is empty.
6. When the user enters age:

- Check whether the value is numeric.
- Verify that the age is within the permitted range.
- 7. When the user enters an email:
 - Check whether it follows a valid email format.
- 8. When the user enters a phone number:
 - Check whether it contains the required number of digits.
- 9. When the user selects a membership plan:
 - Verify that a valid option is selected.
- 10. Display valid/error messages live beside the corresponding fields.
- 11. On form submission, perform validation again for all fields.
- 12. If all fields are valid, display a successful admission message.
- 13. Otherwise, prevent form submission and display appropriate errors.
- 14. Stop

Conclusion:

The Gym Admission Form was successfully designed with event-based live input validation. JavaScript events such as input, blur, change, and submit were used to access and validate form fields dynamically. The system provides immediate feedback to the user and prevents invalid or incomplete information from being submitted. Thus, event-driven form validation improves data accuracy, reliability, and user experience.